

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Winter Examination – 2024

Course: B.Tech.

Branch: Common To All Branches

Semester: I

Subject Code & Name: 24AF2EGRES104; Engineering Graphics

Max Marks: 60

Date: 11/02/2025

Duration: 4 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level /CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 Set squares are primarily used for drawing which angles? A) 30°, 45°, 60°, and 90° B) 15°, 35°, 75°, and 95° C) 10°, 20°, 50°, and 80° D) 25°, 50°, 70°, and 100°	(CO1)	1
2 What is the purpose of a center line in a drawing? A) To show visible outlines B) To represent symmetry and centers of circles C) To indicate section lines D) To show cutting planes	(CO1)	1
3 Which of the following is true for first-angle projection? A) The object is placed between the observer and the projection plane B) The projection plane is placed between the observer and the object C) Views are placed as they appear in reality D) It is the standard method in the United States	(CO2)	1
4 If a point is located on the XY plane in orthographic projection, which of the following is true? A) Its Z-coordinate is zero B) Its X-coordinate is zero C) Its Y-coordinate is zero D) It is in the first quadrant	(CO2)	1
5 The true length of a line is seen in which of the following conditions? A) When the line is parallel to the plane B) When the line is perpendicular to the plane C) When the line is inclined to both HP and VP D) When viewed from the side	(CO3)	1
6 If a line is located in the first quadrant and is inclined to HP, where will its front view be located? A) Above the XY line B) Below the XY line C) On the XY line D) At the origin	(CO 3)	1

7	A pentagonal plane resting on HP with an edge inclined to VP will have its top view appearing as: A) A pentagon B) A horizontal line C) A distorted pentagon D) A vertical line	(CO 4)	1
8	If a plane is inclined to both HP and VP, its projections appear as: A) A true shape in both views B) A point in both views C) Inclined lines in both views D) A horizontal and vertical line	(CO 4)	1
9	When a prism is lying on HP with its axis inclined to HP, its front view will be: A) A rectangle B) A parallelogram C) A line D) An ellipse	(CO 4)	1
10	Which of the following is NOT a principal plane in orthographic projection? A) Horizontal plane (HP) B) Vertical plane (VP) C) Side plane (SP) D) Profile plane (PP)	(CO 5)	1
11	The angles between the projection of the x-axis, y-axis, and z-axis in an isometric view are: A) 90° B) 120° C) 45° D) 60°	(CO 5)	1
12	When a solid is cut by a plane, the shape of the section depends on: A) The position of the plane B) The material of the solid C) The color of the solid D) The surface texture of the solid	(CO 5)	1
Q.2 Solve the following.			12
A)	Draw the following lines by stating their description and general applications: i. Continuous thick or Continuous wide ii. Dashed thin (narrow) iii. Chain thin Long-dashed dotted (narrow)	(CO 1)	6
B)	Differentiate Aligned and Uni-directional system of placing the dimensions on a drawing with the help of diagrams.	(CO 2)	6
Q.3 Solve the following.			12
A)	A line AB, 50 mm long, is inclined to the HP at 30° and parallel to the VP. The end nearest to the HP is 40 mm above it and 25 mm in front of the VP. Draw the projections.	(CO 3)	6
B)	A point P is in the first quadrant. Its shortest distance from the intersection point of H.P., V.P. and Auxiliary vertical plane, perpendicular to the H.P. and V.P. is 70 mm and it is equidistant from principal planes (H.P. and V.P.). Draw the projections of the point and determine its distance from the H.P. and V.P.	(CO 2)	6
Q.4 Solve Any Two of the following.			12
A)	Draw the projections of a regular hexagon of 25 mm side, having one of its sides in the H.P. and inclined at 60° to the V.P., and its surface making an angle of 45° with the H.P.	(CO 4)	6

- B) A square pyramid of side of base 40 mm and length of axis 60 mm is resting on its corner of base on ground with an edge of the base through that corner making an angle of 60° with the HP. The apex is away from the observer and the axis is parallel to the HP. Draw the projections if the axis is inclined to the VP at 20° . (CO 4) 6
- C) A cone of diameter 60 mm and height 60 mm is resting on the HP on one of its generators. Draw its projections if its axis is parallel to the VP. (CO 4) 6

Q.5 Solve Any Two of the following.

- A) A pentagonal pyramid having a base side of 45 mm and a slant length of 80 mm rests on its base on the HP with a base edge AB perpendicular to the VP. A section plane passing through corner D and perpendicular to the slant face ABO cuts the solid. Draw FV and sectional TV. (CO 5) 6
- B) A cylinder of 40 mm diameter, 60 mm height and having its axis vertical, is cut by a section plane, perpendicular to the V.P., inclined at 45° to the H.P. and intersecting the axis 32 mm above the base. Draw its front view, sectional top view, sectional side view and true shape of the section (CO 5) 6
- C) Draw the development of the lateral surface of the part P of the triangular pyramid as shown in Fig. 1. The line $o'1'$ in the front view is the true length of the slant edge because it is parallel to xy in the top view. The true length of the side of the base is seen in the top view. (CO 6) 6

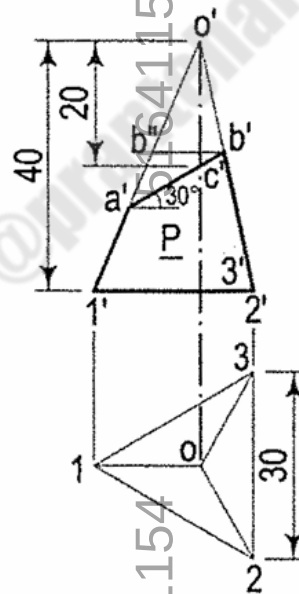


Fig. 1

Q.6 Solve Any Two of the following.

- A) Draw the FV and TV of the object shown in Fig. 2 using the third-angle method. (CO 5) 6

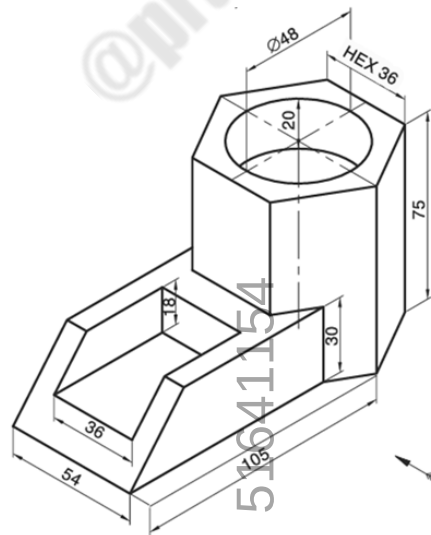


Fig. 2

- B) Draw the half-sectional FV, and half-sectional RHSV of the object as shown in Fig.3 (CO 5) 6
by using first angle method.

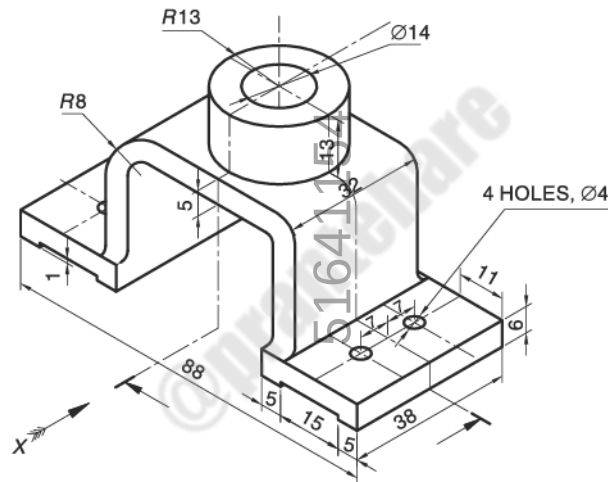


Fig. 3

- C) Fig. 4 shows the FV and LHSV of an object. Draw its isometric view. (CO 5) 6

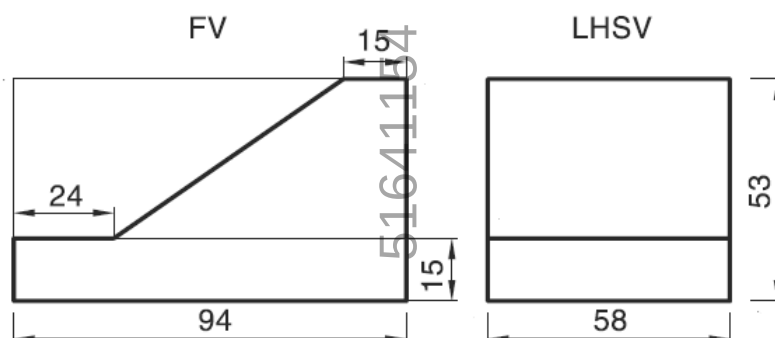


Fig. 4

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